

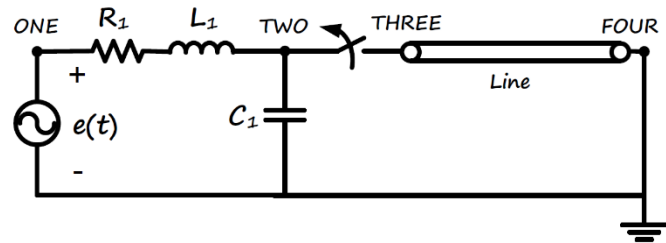
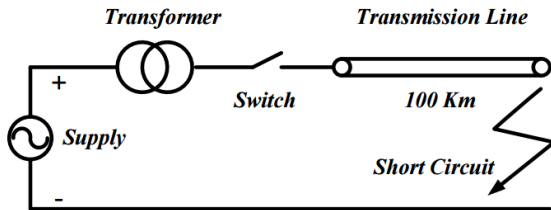
Fall 2025, EE 582: Assignment 3:

Nodal-Analysis-based Modeling of Electric Circuits

Due on Sunday, October 19 (submission in Canvas)

SECTION A (Transmission Line Interruption)

This section of the assignment is similar to Section-C of Assignment#2, with the difference that the transmission line is to be modeled based on the traveling-wave concept rather than using a π circuit that was assumed in Assignment#2.

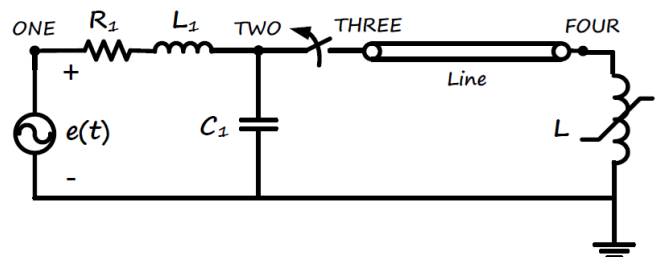
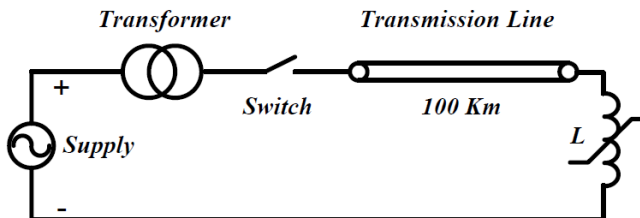


We have again assumed a single-phase circuit here for simplicity and that the fault has already been applied (i.e., node Four is short-circuited due to the fault). The parameters of the circuit are: $e(t) = 230 \times \cos(\omega t)$ kV, $\omega = 377$ rad/s, $R_1 = 3 \Omega$, $L_1 = 350$ mH, $C_1 = 10$ nF. The transmission line also has the following parameters: $R = 0.5 \Omega/\text{km}$, $L = 1$ mH/km, $C = 12$ nF/km. Assume that the system is operating with zero initial conditions, and the breaker is initially closed. At $t = 8$ ms, the breaker is commanded to open the line, but it will not actually open until the current passes through zero. The study period is from $t = 0$ to $t = 20$ ms. Use time-step size $\Delta t = 20 \mu\text{s}$ for PSCAD simulations and your MATLAB codes.

- 1) Case-1: Obtain the solution from PSCAD using transmission line model.
- 2) Case-2: Assume the line is split into two segments and the line resistance is lumped 1/4 at each end and 1/2 in the middle. Write your own MATLAB code assuming the Trapezoidal integration rule to solve the system.
- 3) Case-3: Assume the line is split into two segments and the line resistance is lumped 1/2 at each end. Write your own MATLAB code assuming Trapezoidal integration rule to solve the system.
- 4) Show the results for voltages at nodes *Two* and *Three* and the fault current (i.e., from node *Four* to ground) in three separate (big enough) plots. In each plot, superimpose the results of the three Cases. Comment on the accuracy of your code results compared to those of PSCAD.

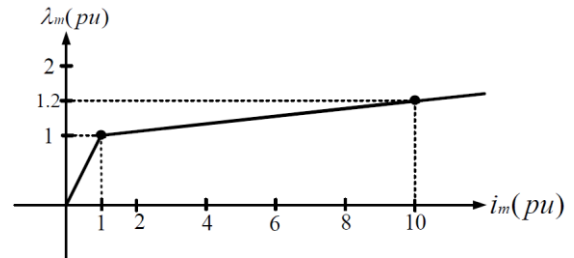
SECTION B (Transmission Line Interruption)

This section is similar to Section-A, with the difference that the transmission line is terminated to a load that can be represented as a nonlinear inductance.



The nonlinear characteristic of the inductance is shown below. For per unit values, we have assumed the peak values of the magnetizing flux and current as the bases: $\lambda_m^{base} = 610 \text{ V.s}$ and $i_m^{base} = 5 \text{ A}$.

All other system conditions and parameters are similar to Section A. The study period is from $t=0$ to $t=20 \text{ ms}$. Use a time-step size of $\Delta t=20 \mu\text{s}$ for PSCAD simulations and your MATLAB code.



- 1) Case-1: Obtain the solution from PSCAD.
- 2) Case-2: Use the transmission line model in Section A that provided more accurate results compared to PSCAD. Write your own MATLAB code assuming the Trapezoidal integration rule to solve the system.
- 3) Show the results for voltages at nodes *Two*, node *Three*, the load current, and the magnetizing current i_m in four separate (big enough) plots. In each plot, superimpose the results of your code and PSCAD. Comment on the accuracy of your code.

SECTION C (Frequency Response of Discretization Rules)

A way to gain insight into the behavior of different discretization rules is to look at the distortion in magnitude and phase angle introduced by the rule on the representation of a simple first-order system like an inductor or a capacitor as a function of frequency.

- 1) Similar to what we did in class for an inductor, derive the expressions for the magnitude and phase distortion of Backward Euler and Trapezoidal rules when applied to discretize a capacitor; and plot them as functions of f_{pu} from 0 to 0.5.
- 2) In the class, we found an equivalent circuit for an inductor L when it was discretized by Trapezoidal rule, which turned out to be a frequency-dependent inductor L_d . Find equivalent circuits for an inductor L and also a capacitor C , when discretized using Backward Euler rule.

REPORTING

All reports should be typed. The assignment report should demonstrate the individual's work and include printouts of the model windows, simulation results, as well as answers, discussions, and conclusions wherever appropriate. Use graphical commands such as **hold** and **subplot** for compact and clear presentation of the figures in your report. The discussions and comments should not be long (**avoid writing unnecessarily long reports!**), but they should demonstrate your understanding of the problem.

Please provide your MATLAB codes as part of your report.

Best of luck!